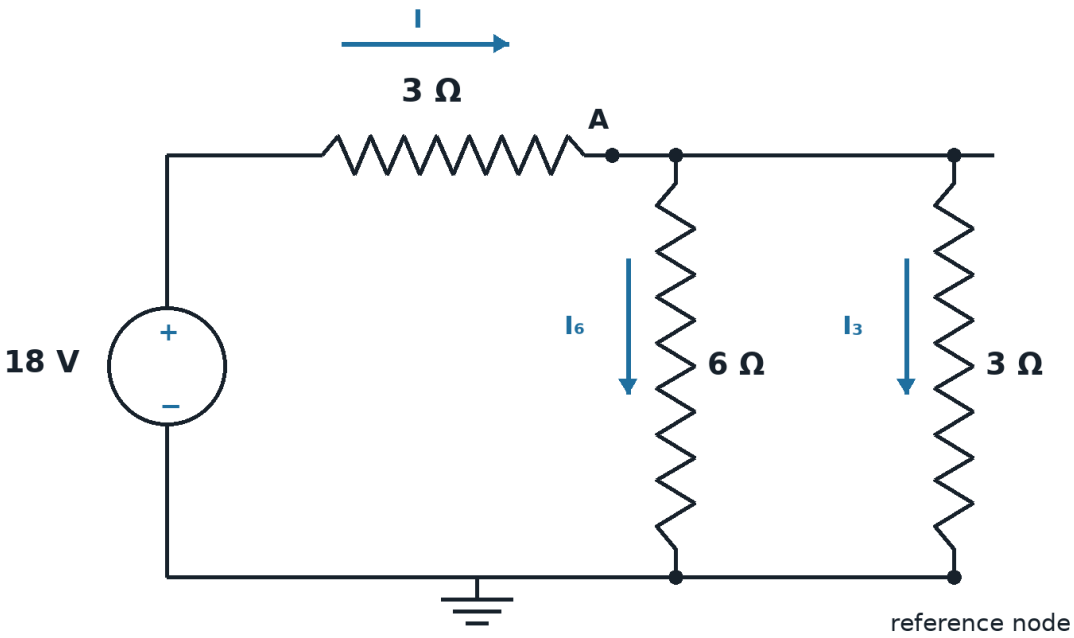
KVL-KCL Questions

Question 1

Find the series current and both downward branch currents.



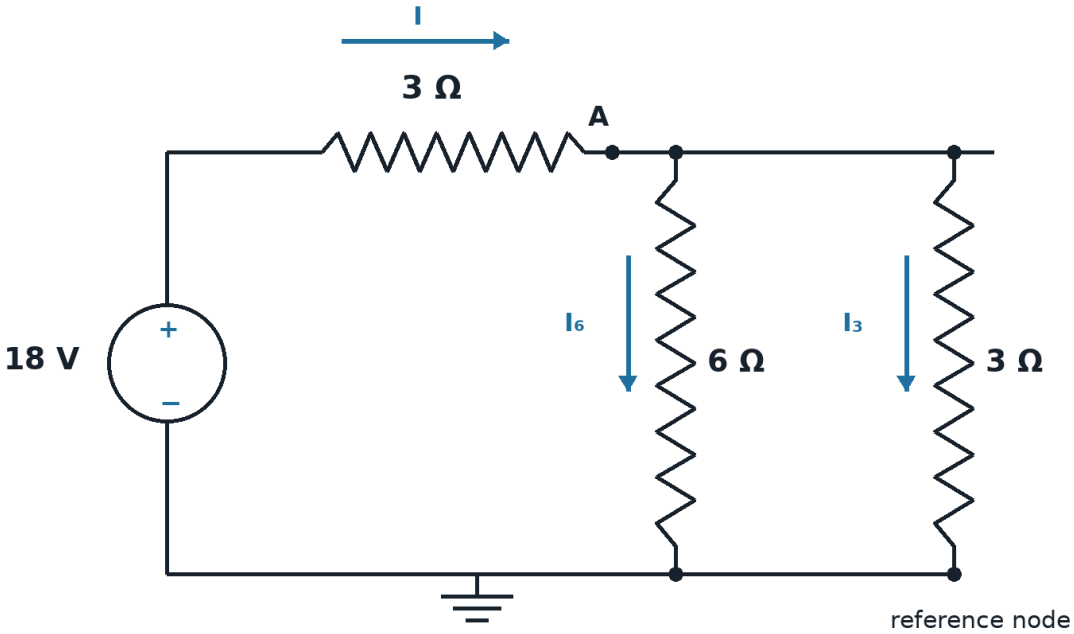
An 18 V source is positive at the top. Its top terminal connects through $3\ \Omega$ to node A. Resistors of $6\ \Omega$ and $3\ \Omega$ connect from A to the bottom reference node. The series arrow points right; both branch arrows point downward.

Solution

Let node A be V_A and take the labeled directions as positive. KVL gives $18 - 3I - V_A = 0$. KCL and Ohm's law give $I = I_6 + I_3 = V_A/6 + V_A/3 = V_A/2$. Hence $18 - 3(V_A/2) - V_A = 0$, so $V_A = 7.20$ V. Therefore $I = 3.60$ A right, $I_6 = 1.20$ A downward, and $I_3 = 2.40$ A downward.

Question 2

Write one node equation for A and use it to check KCL numerically.



An 18 V source is positive at the top. Its top terminal connects through $3\ \Omega$ to node A. Resistors of $6\ \Omega$ and $3\ \Omega$ connect from A to the bottom reference node. The series arrow points right; both branch arrows point downward.

Solution

With currents leaving A positive, $

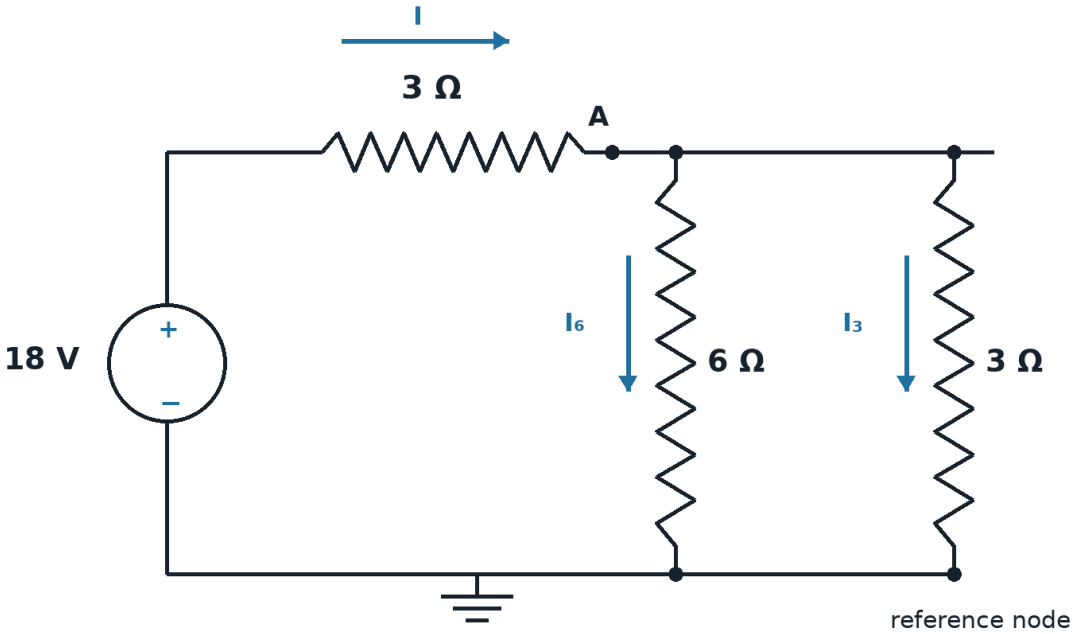
rac{V_A-18}{3}+

rac{V_A}{6}+

rac{V_A}{3}=0.$Multiplying by 6 gives 2V_A-36+V_A+2V_A=0,$ so V_A=7.20\ \mathrm{V}.$Numerically, currer
7.2)/3=3.60\ \mathrm{A}$ and current leaving is $7.2/6+7.2/3=3.60\ \mathrm{A}$.

Question 3

Find the power of every element and verify conservation of power.



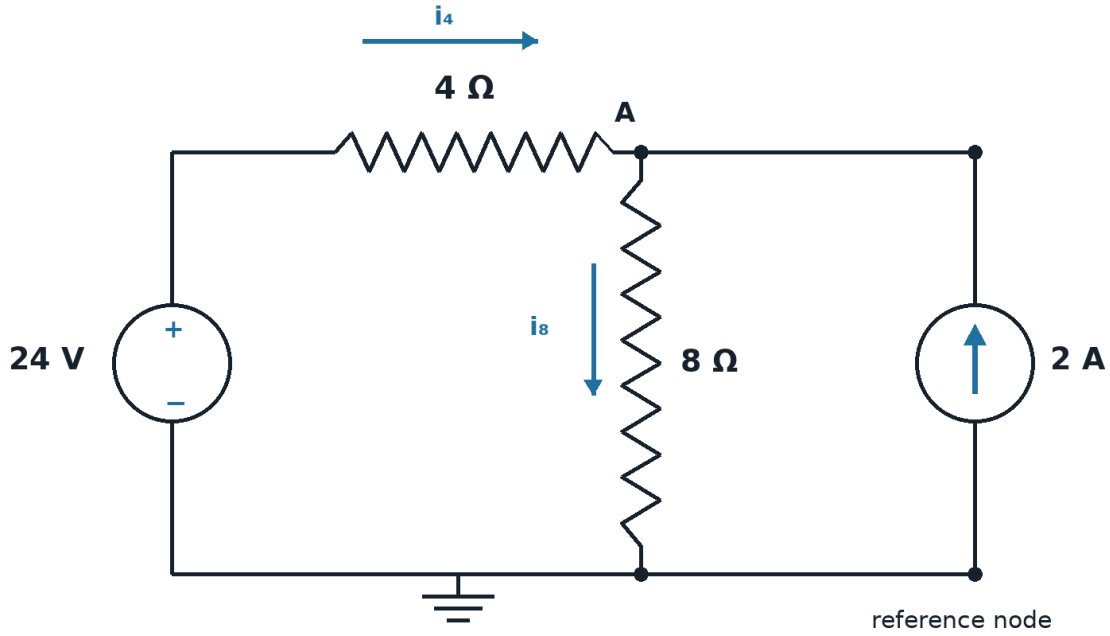
An 18 V source is positive at the top. Its top terminal connects through $3\ \Omega$ to node A. Resistors of $6\ \Omega$ and $3\ \Omega$ connect from A to the bottom reference node. The series arrow points right; both branch arrows point downward.

Solution

From KVL and KCL, $I = 3.60$ A, $I_6 = 1.20$ A, and $I_3 = 2.40$ A. With positive power defined as absorbed, $P_{3\Omega,\text{series}} = (3.6)^2(3) = 38.88$ W, $P_{6\Omega} = (1.2)^2(6) = 8.64$ W, and $P_{3\Omega,\text{branch}} = (2.4)^2(3) = 17.28$ W. Source current leaves its positive terminal, so $P_{18\text{V}} = -(18)(3.6) = -64.80$ W; it delivers 64.80 W. The check is $38.88 + 8.64 + 17.28 - 64.80 = 0$ W.

Question 4

Find node voltage A and the two resistor currents.



A 24 V *source is positive at the top and fixes the left node. A* $4\ \Omega$ *resistor connects it to node A. An* $8\ \Omega$ *resistor connects A to reference. A* 2 A *source points upward from reference to A. Resistor arrows point right through* $4\ \Omega$ *and downward through* $8\ \Omega$*.*

Solution

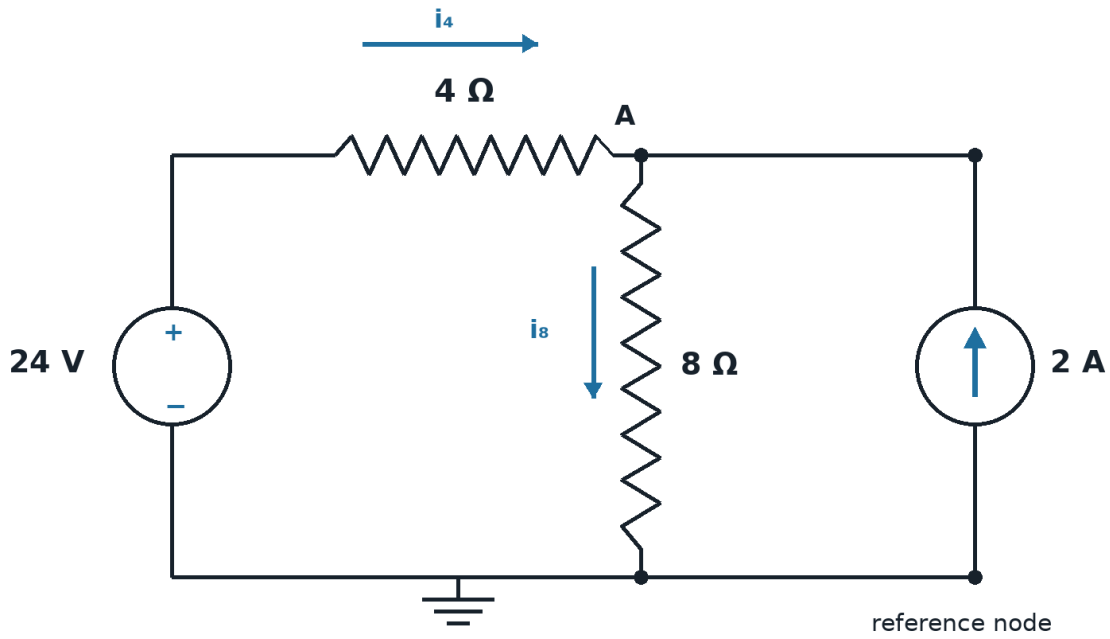
Take currents leaving A as positive. KCL gives $

rac{V_A-24}{4}+

rac{V_A}{8}-2=0.$Thus$2V_A-48+V_A-16=0andV_A=64/3=21.33\ \mathrm{V}.$Therefore$i_4=(24-V_A)/4=2/3\ \mathrm{A}$rightand$i_8=V_A/8=8/3\ \mathrm{A}$ downward.

Question 5

Classify each source as absorbing or delivering power and verify the power balance.



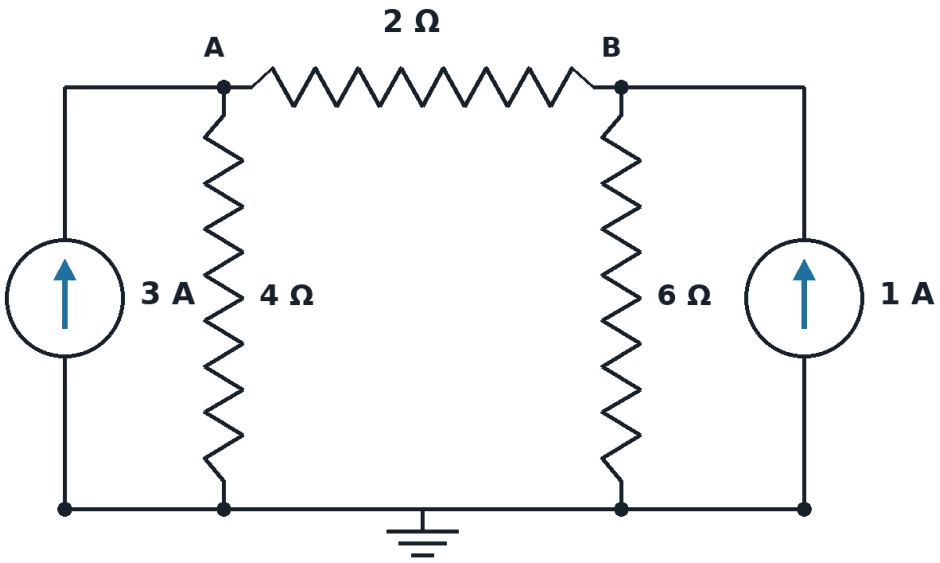
A 24 V *source is positive at the top and fixes the left node. A* $4\ \Omega$ *resistor connects it to node A. An* $8\ \Omega$ *resistor connects A to reference. A* 2 A *source points upward from reference to A. Resistor arrows point right through* $4\ \Omega$ *and downward through* $8\ \Omega$*.*

Solution

Using $V_A = 64/3$ V, the resistors absorb $P_{4\Omega} = (2/3)^2(4) = 16/9$ W and $P_{8\Omega} = (8/3)^2(8) = 512/9$ W. The voltage-source current leaves its positive terminal, so $P_{24\text{V}} = -24(2/3) = -16$ W; it delivers. The current source enters its negative terminal, so $P_{2\text{A}} = -(64/3)(2) = -128/3$ W; it delivers. The check is $16/9 + 512/9 - 16 - 128/3 = 0$ W.

Question 6

Use nodal analysis to find both node voltages and the current through the linking resistor.



Nodes A and B share a $2\ \Omega$ resistor. A has a $4\ \Omega$ resistor to reference and a 3 A source pointing upward from reference. B has a $6\ \Omega$ resistor to reference and a 1 A source pointing upward from reference. The bottom wire is reference.

Solution

Take currents leaving each node as positive. The equations are $

rac{V_A}{4}+

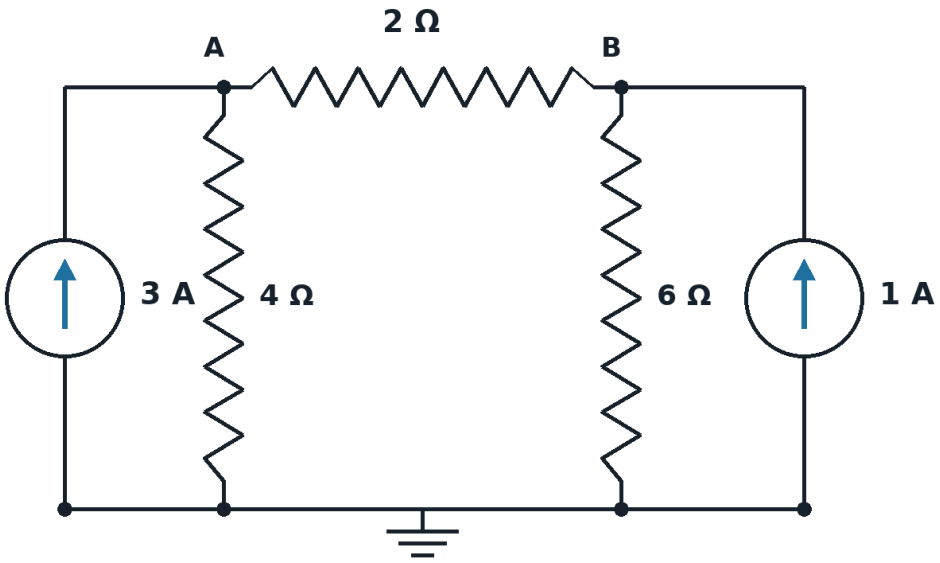
rac{V_A-V_B}{2}-3=0and

rac{V_B}{6}+

rac{V_B-V_A}{2}-1=0.$Equivalently$,3V_A-2V_B=12and-3V_A+4V_B=6.$Solvinggives$V_A=10.0\ \mathrm{V}andV_B=9.0\ \mathrm{V}.$Thelinkcurrentis$(V_A-V_B)/2=0.500\ \mathrm{A}$ from A to B.

Question 7

Find the power status of both current sources and check total power.



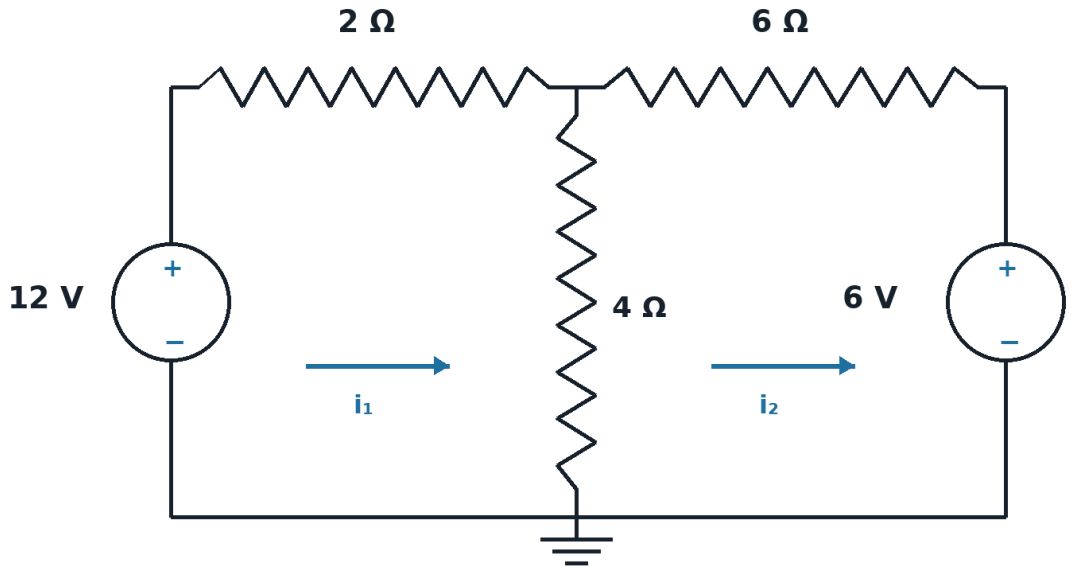
Nodes A and B share a $2\ \Omega$ resistor. A has a $4\ \Omega$ resistor to reference and a 3 A source pointing upward from reference. B has a $6\ \Omega$ resistor to reference and a 1 A source pointing upward from reference. The bottom wire is reference.

Solution

From the node equations, $V_A = 10$ V and $V_B = 9$ V. Both source currents enter their negative terminals. Thus $P_{3\text{A}} = -(10)(3) = -30$ W and $P_{1\text{A}} = -(9)(1) = -9$ W; both deliver. Resistor absorption is $10^2/4 + (10-9)^2/2 + 9^2/6 = 25 + 0.5 + 13.5 = 39$ W. Therefore $39 - 30 - 9 = 0$ W.

Question 8

Define clockwise mesh currents and solve for them.



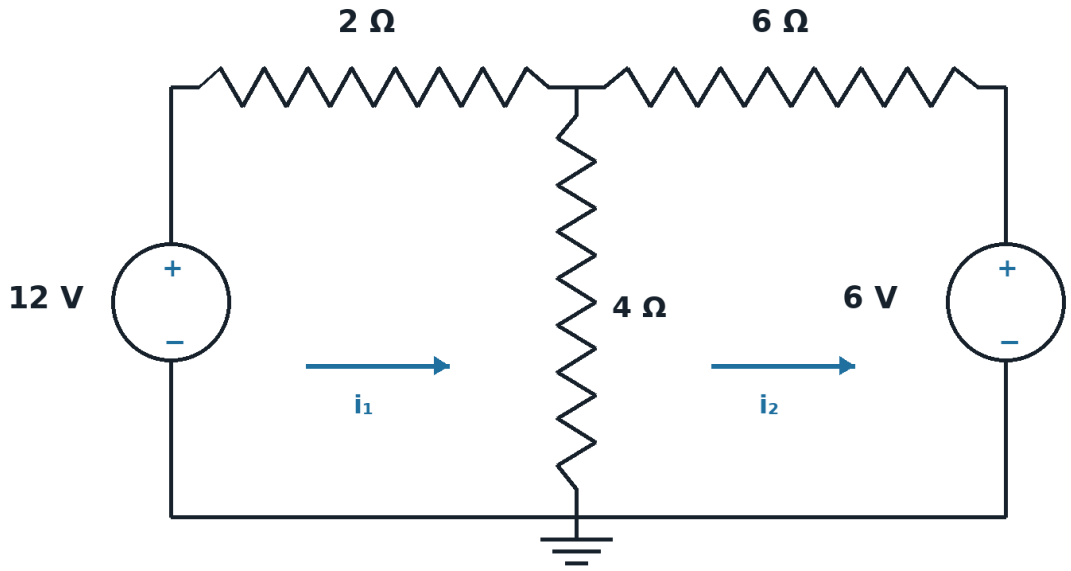
Two clockwise mesh currents are shown. The left loop has a 12 V *source positive at the top and a top* $2\ \Omega$ *resistor. The right loop has a top* $6\ \Omega$ *resistor and a* 6 V *source positive at the top. A shared vertical* $4\ \Omega$ *resistor separates the loops; the bottom wire is common.*

Solution

Let i_1 and i_2 be clockwise in the left and right meshes. KVL gives $-12 + 2i_1 + 4(i_1 - i_2) = 0$ and $6i_2 + 6 + 4(i_2 - i_1) = 0$. Thus $6i_1 - 4i_2 = 12$ and $-4i_1 + 10i_2 = -6$. Solving gives $i_1 = 24/11 = 2.18$ A clockwise and $i_2 = 3/11 = 0.273$ A clockwise.

Question 9

Find the current in the shared resistor and state its physical direction.



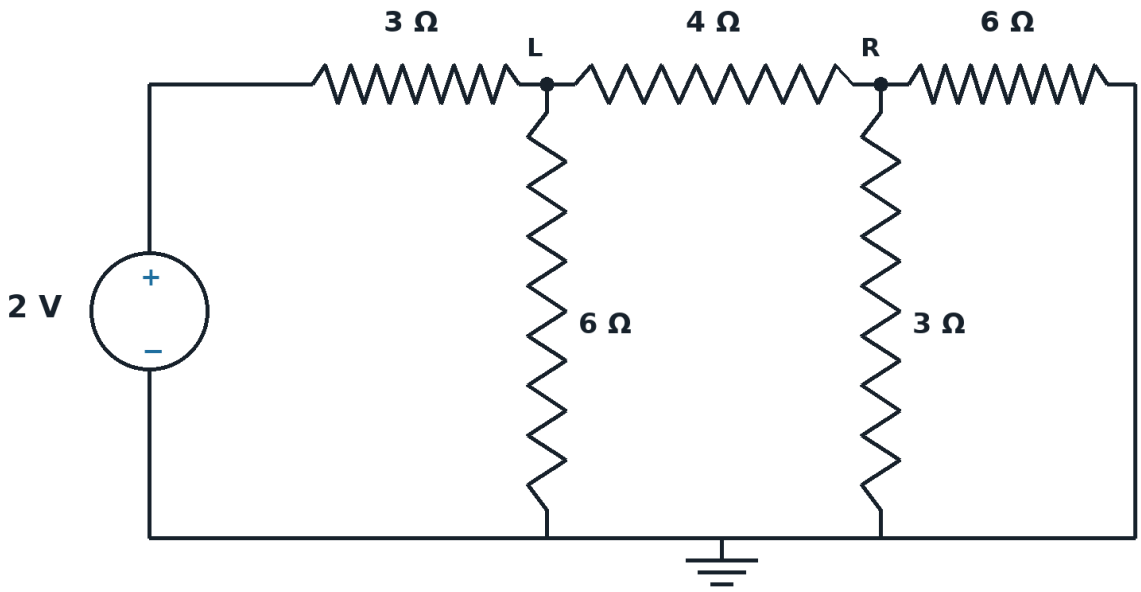
Two clockwise mesh currents are shown. The left loop has a 12 V *source positive at the top and a top* $2\ \Omega$ *resistor. The right loop has a top* $6\ \Omega$ *resistor and a* 6 V *source positive at the top. A shared vertical* $4\ \Omega$ *resistor separates the loops; the bottom wire is common.*

Solution

The mesh equations are $6i_1 - 4i_2 = 12$ and $-4i_1 + 10i_2 = -6$, which give $i_1 = 24/11$ A and $i_2 = 3/11$ A. The shared-branch current, downward as referenced by the left mesh, is $i_1 - i_2 = 21/11 = 1.91$ A. It is positive, so the physical direction is downward.

Question 10

Use nodal analysis to find the voltages at L and R.



A 12 V *source fixes the left top node. Along the top are* $3\ \Omega$ *to L,* $4\ \Omega$ *from L to R, and* $6\ \Omega$ *from R to the reference return. L also has* $6\ \Omega$ *to reference, and R has* $3\ \Omega$ *to reference.*

Solution

With the bottom wire as reference, KCL gives $

rac{V_L-12}{3}+

rac{V_L}{6}+

rac{V_L-V_R}{4}=0and

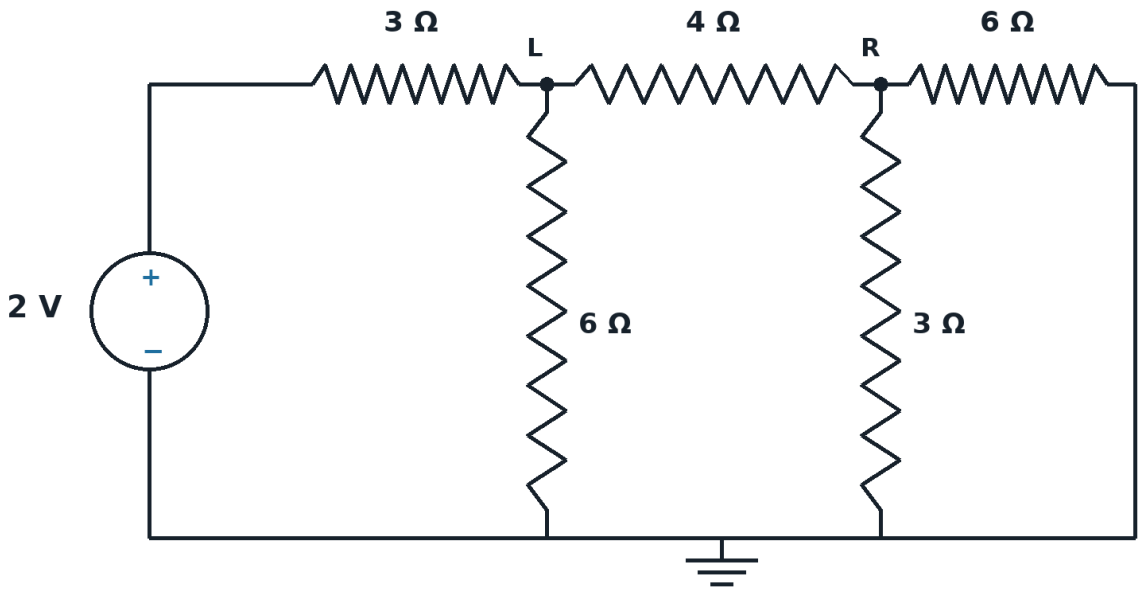
rac{V_R-12}{6}+

rac{V_R}{3}+

rac{V_R-V_L}{4}=0.$Multiplyingby$12$gives$9V_L-3V_R=48and-3V_L+9V_R=24.$Solvinggives$V_L=7.00\ \mathrm{V}andV_R=5.00\ \mathrm{V}$.

Question 11

Find the magnitude and direction of every resistor current.



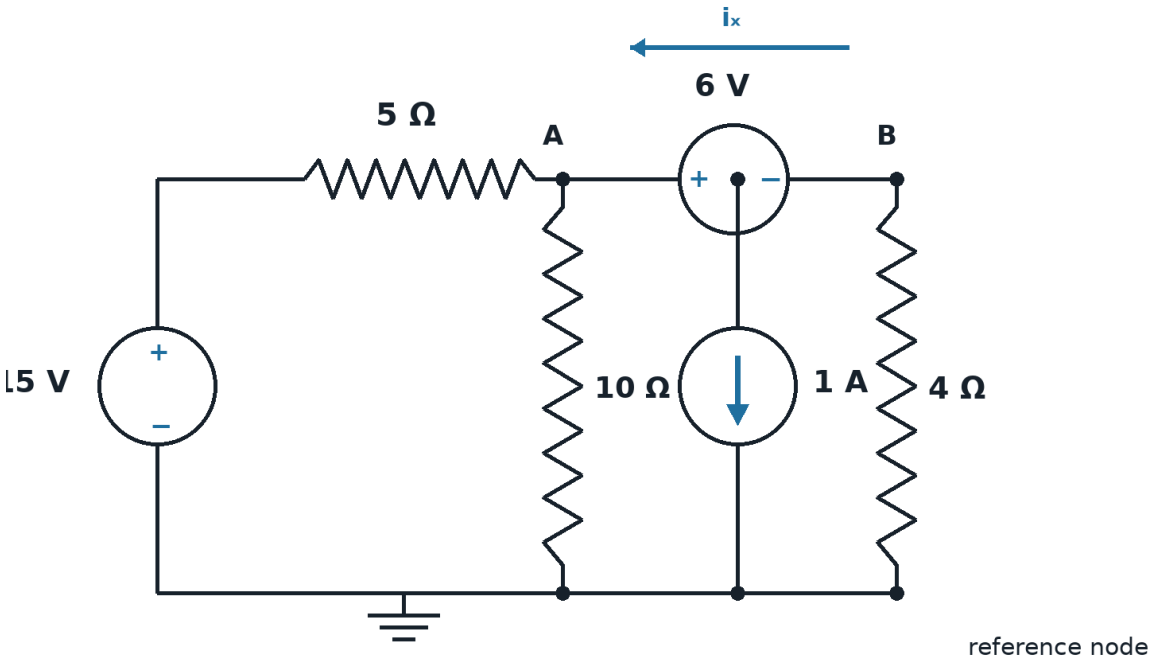
A 12 V source fixes the left top node. Along the top are $3\ \Omega$ to L, $4\ \Omega$ from L to R, and $6\ \Omega$ from R to the reference return. L also has $6\ \Omega$ to reference, and R has $3\ \Omega$ to reference.

Solution

Using $V_L = 7$ V and $V_R = 5$ V, the currents are $(12-7)/3 = 5/3$ A from the source node to L, $7/6$ A from L to reference, $(7-5)/4 = 0.500$ A from L to R, $5/3$ A from R to reference through $3\ \Omega$, and $(12-5)/6 = 7/6$ A from the source node to R. KCL checks at each node.

Question 12

Form a supernode and find the two node voltages.



A 15 V *source feeds A through* $5\ \Omega$*. From A to reference are* $10\ \Omega$ *and a downward* 1 A *source. A* 6 V *source connects A, its positive terminal, to B, its negative terminal. B has* $4\ \Omega$ *to reference. The reference arrow through the* 6 V *source points from B to A.*

Solution

The constraint is $V_A - V_B = 6$. KCL for the supernode is $

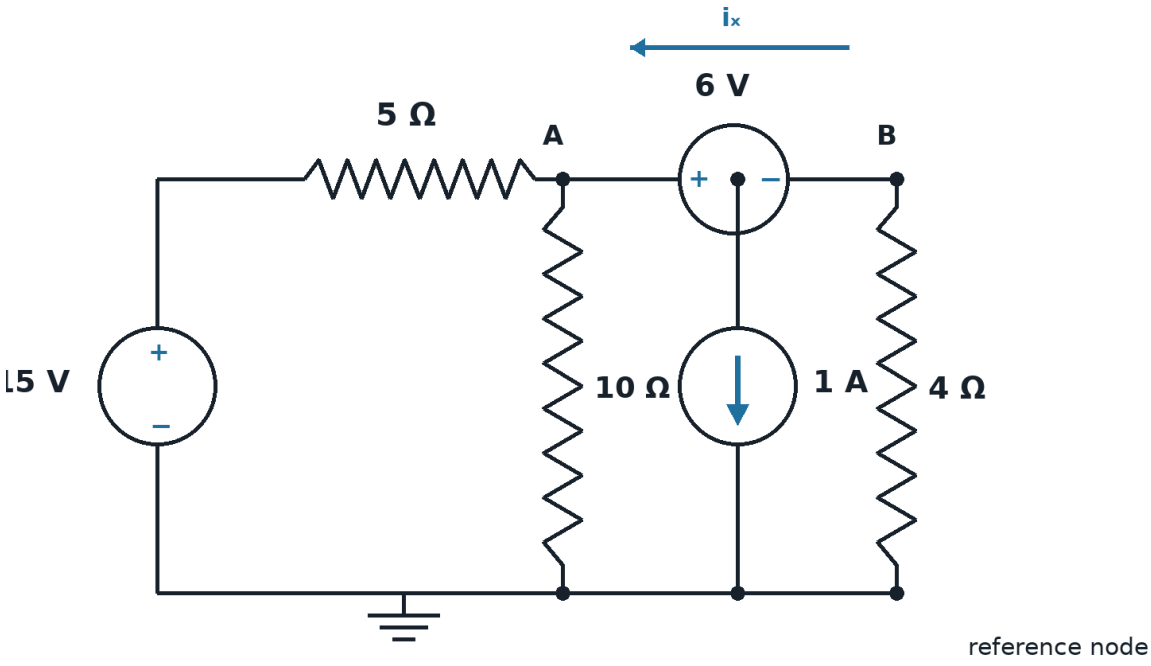
rac{V_A-15}{5}+

rac{V_A}{10}+1+

rac{V_B}{4}=0.$Multiplying by 20 gives 6V_A+5V_B=40.With V_B=V_A-6, 11V_A=70.Therefore V_A=70/11=6
\mathrm{V} and V_B=4/11=0.364\ \mathrm{V}$.

Question 13

Find the referenced current through the six-volt source, interpret its sign, and classify its power.



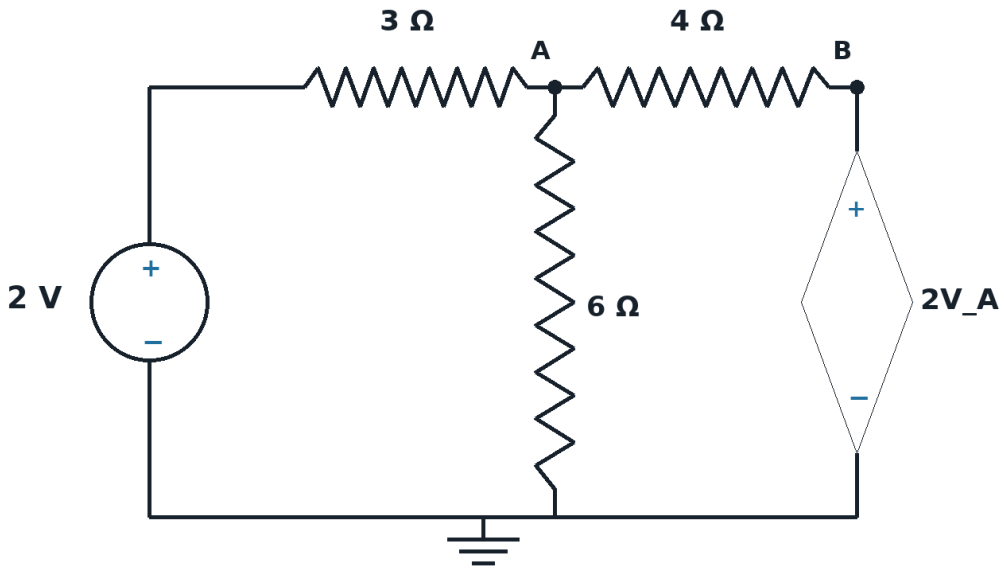
A 15 V *source feeds A through* $5\ \Omega$*. From A to reference are* $10\ \Omega$ *and a downward* 1 A *source. A* 6 V *source connects A, its positive terminal, to B, its negative terminal. B has* $4\ \Omega$ *to reference. The reference arrow through the* 6 V *source points from B to A.*

Solution

From supernode analysis, $V_A = 70/11$ V and $V_B = 4/11$ V. KCL at B gives $i_x + V_B/4 = 0$, so $i_x = -1/11$ A $= -90.9$ mA. The actual current is 90.9 mA from A to B, opposite the arrow. It enters the positive terminal, so $P_{6\text{V}} = 6(1/11) = 6/11 = 0.545$ W absorbed.

Question 14

Use the dependent-source constraint and KCL to find both node voltages.



A 12 V *source feeds node A through* $3\ \Omega$*. A has* $6\ \Omega$ *to reference and connects to B through* $4\ \Omega$*. A dependent voltage source connects B to reference and enforces* $V_B = 2V_A$*, positive at B.*

Solution

The source constraint is $V_B = 2V_A$. KCL at A gives $

rac{V_A-12}{3}+

rac{V_A}{6}+

rac{V_A-V_B}{4}=0.$Substitutetheconstraint$:

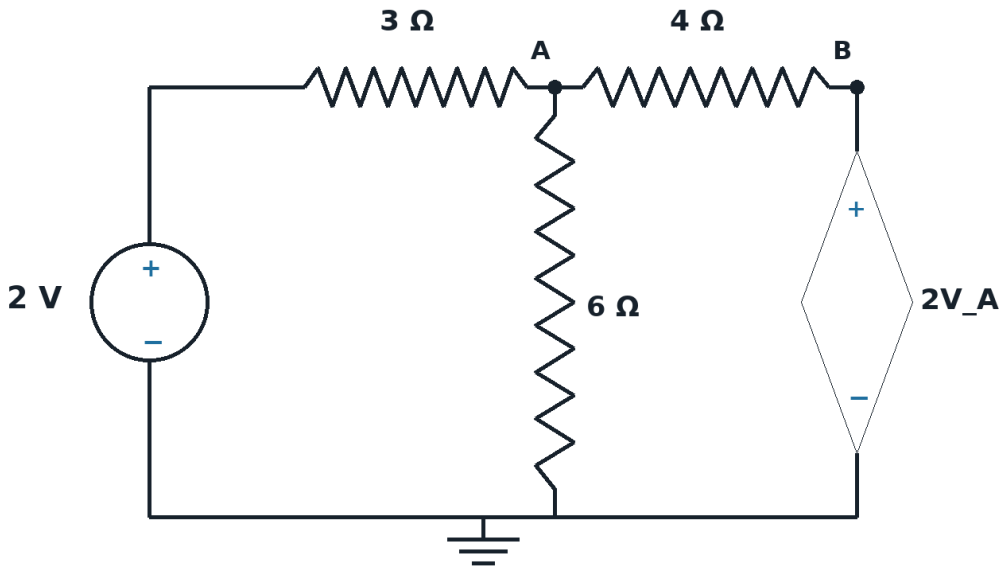
rac{V_A-12}{3}+

rac{V_A}{6}-

rac{V_A}{4}=0.$Multiplyingby$12$gives$4V_A-48+2V_A-3V_A=0.$Hence$V_A=16.0\ \mathrm{V}andV_B=32.0\ \mathrm{V}$.

Question 15

Find the dependent voltage-source current and verify the circuit power balance.



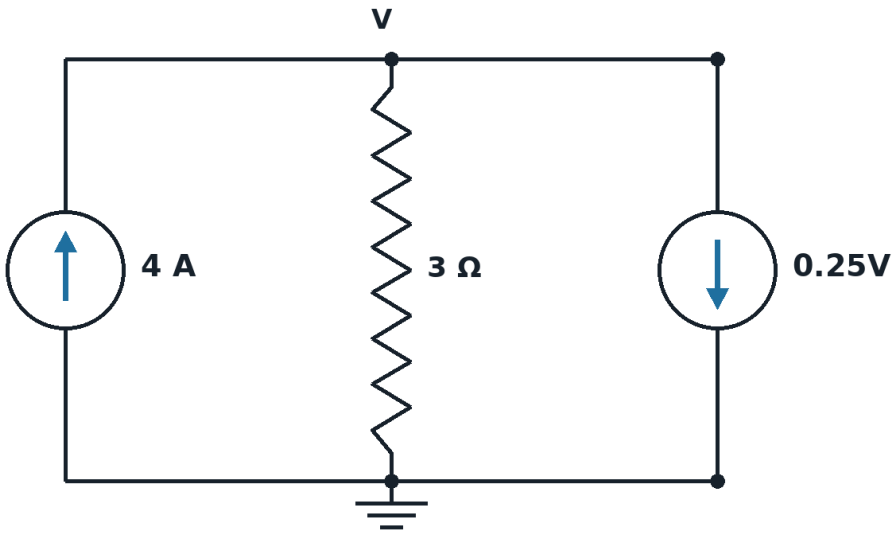
A 12 V *source feeds node A through* $3\ \Omega$*. A has* $6\ \Omega$ *to reference and connects to B through* $4\ \Omega$*. A dependent voltage source connects B to reference and enforces* $V_B = 2V_A$*, positive at B.*

Solution

Using $V_A = 16$ V and $V_B = 32$ V, KCL at B gives downward source current $i_d = -(V_B - V_A)/4 = -4.00$ A. Thus the actual current is upward and the dependent source delivers $-(32)(4) = -128$ W. The independent source absorbs $12(4/3) = 16$ W. The resistors absorb $(4/3)^2(3) = 16/3$ W, $16^2/6 = 128/3$ W, and $4^2(4) = 64$ W. The check is $16 + 16/3 + 128/3 + 64 - 128 = 0$ W.

Question 16

Find the node voltage and both downward branch currents.



Three parallel branches connect node V to reference: an independent 4 A *source points upward, a* $3\ \Omega$ *resistor, and a dependent current source of* $0.25V$ *amperes points downward. Voltage V is measured from the top node to reference.*

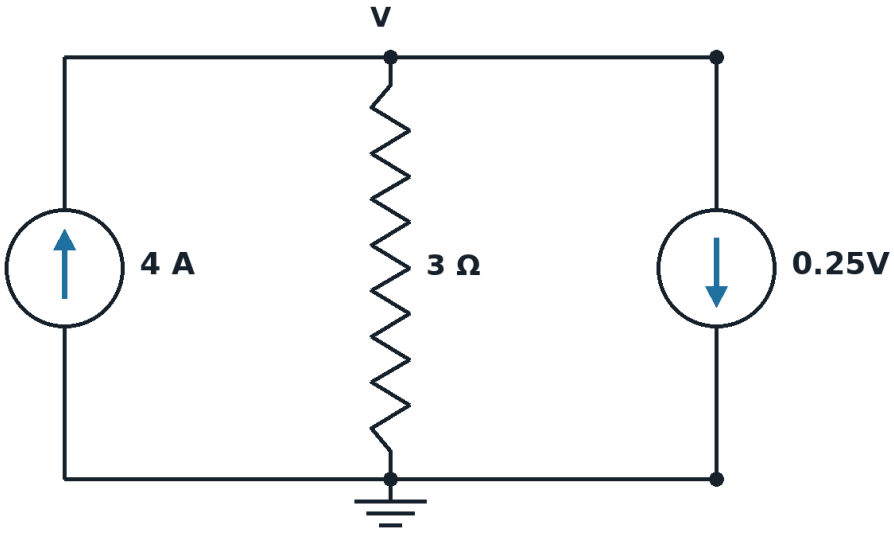
Solution

Take downward currents from node V as positive. KCL gives $

rac{V}{3}+0.25V-4=0.$Thus$(1/3+1/4)V=4andV=48/7=6.86\\mathrm{V}.$Theresistorcurrentis$V/3=16/7=
\mathrm{A}$downward.Thedependent−sourcecurrentis$0.25V=12/7=1.71\\mathrm{A}$downward.Theirsumis$4
\mathrm{A}$.

Question 17

Find the power of all three elements and verify conservation of power.



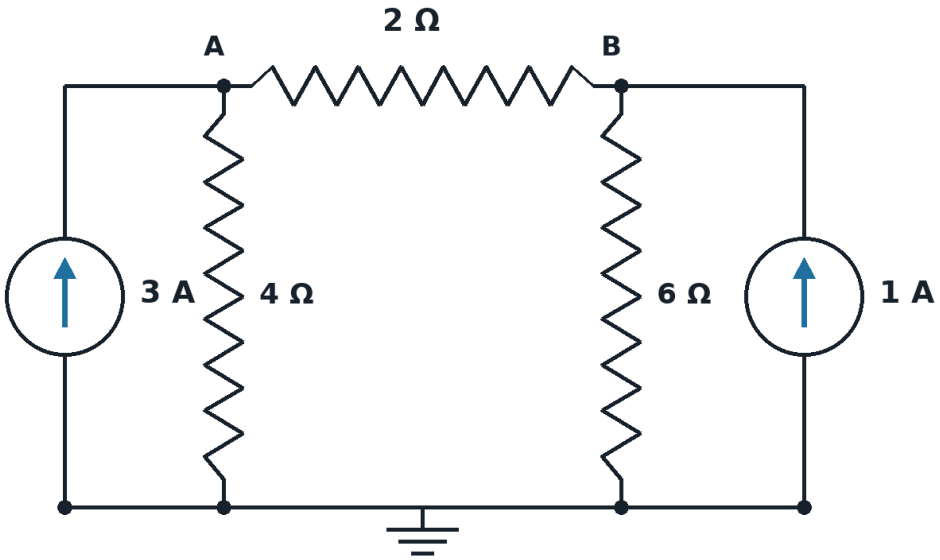
Three parallel branches connect node V to reference: an independent 4 A *source points upward, a* $3\ \Omega$ *resistor, and a dependent current source of* $0.25V$ *amperes points downward. Voltage V is measured from the top node to reference.*

Solution

Using $V = 48/7$ V, the independent source current enters its negative terminal, so $P_{4\text{A}} = -(48/7)(4) = -192/7 = -27.43$ W; it delivers. The resistor absorbs $P_R = V^2/3 = 768/49 = 15.67$ W. The dependent source current enters its positive terminal, so $P_d = V(0.25V) = 576/49 = 11.76$ W absorbed. The check is $768/49 + 576/49 - 192/7 = 0$ W.

Question 18

A student claims that current in the linking resistor is 0.500 A from B to A. Test the claim using KCL.



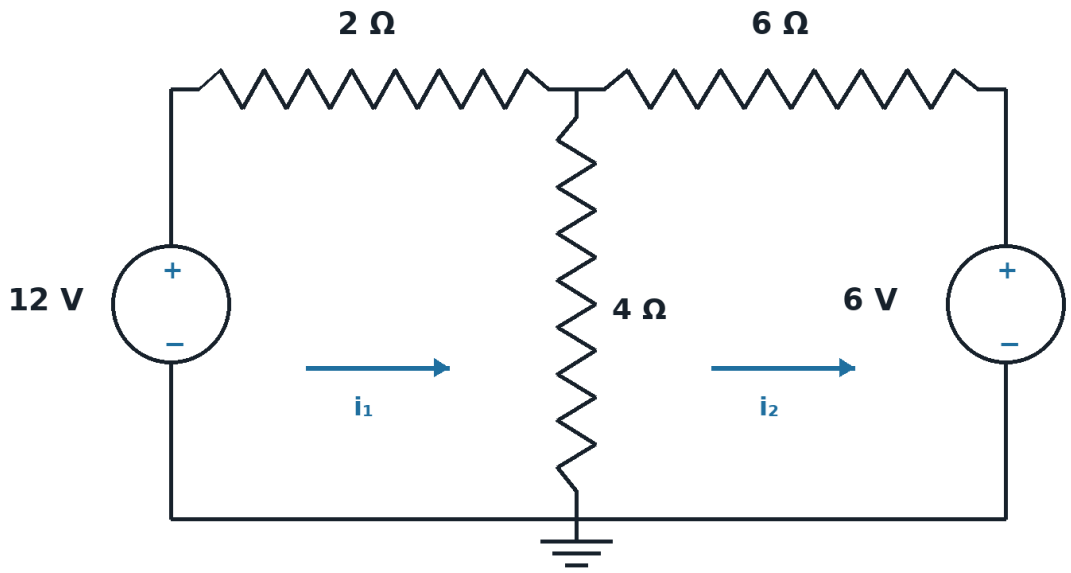
Nodes A and B share a $2\ \Omega$ resistor. A has a $4\ \Omega$ resistor to reference and a 3 A source pointing upward from reference. B has a $6\ \Omega$ resistor to reference and a 1 A source pointing upward from reference. The bottom wire is reference.

Solution

The node equations give $V_A = 10$ V and $V_B = 9$ V. Defining link current from B to A gives $i_{BA} = (V_B - V_A)/2 = -0.500$ A. Therefore the claim is false. The negative sign means the actual current is 0.500 A from A to B. At A, $3 = 10/4 + 0.5 = 3$ A, which confirms KCL.

Question 19

Without resolving the circuit, explain how reversing both mesh-current reference arrows changes the numerical answers and physical currents.



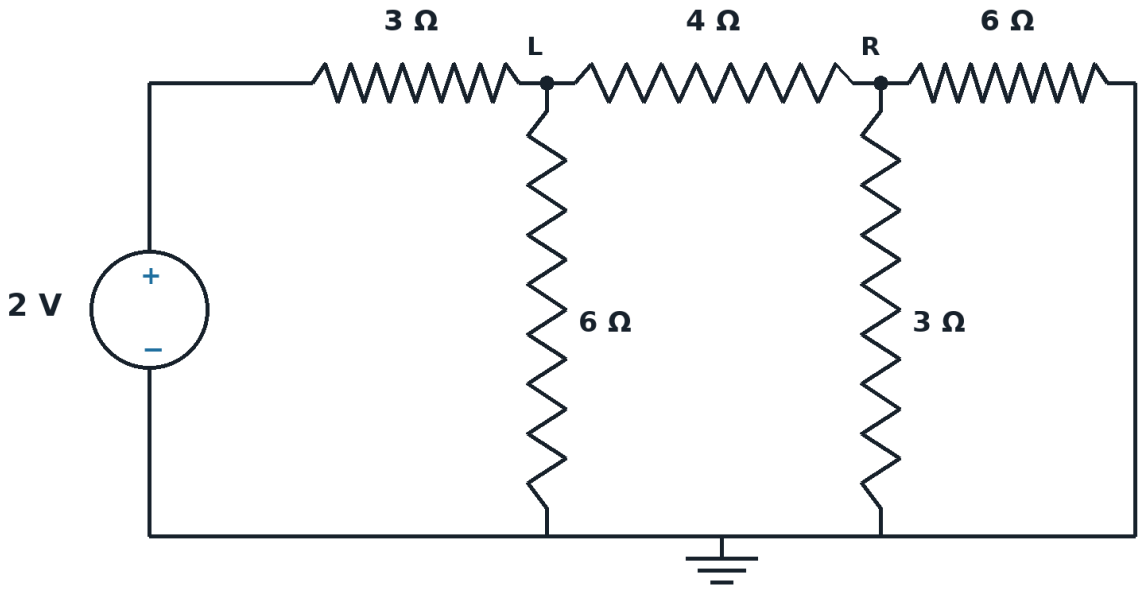
Two clockwise mesh currents are shown. The left loop has a 12 V *source positive at the top and a top* $2\ \Omega$ *resistor. The right loop has a top* $6\ \Omega$ *resistor and a* 6 V *source positive at the top. A shared vertical* $4\ \Omega$ *resistor separates the loops; the bottom wire is common.*

Solution

The original clockwise solutions are $i_1 = 24/11$ A and $i_2 = 3/11$ A. Reversing both reference arrows defines $i_1' = -i_1$ and $i_2' = -i_2$. Thus $i_1' = -24/11$ A and $i_2' = -3/11$ A. Only the reported signs change. The physical branch currents, voltage drops, and powers do not change.

Question 20

A voltmeter reads from R to L. Find its reading, then find the power absorbed by the resistor between L and R.



A 12 V source fixes the left top node. Along the top are $3\ \Omega$ to L, $4\ \Omega$ from L to R, and $6\ \Omega$ from R to the reference return. L also has $6\ \Omega$ to reference, and R has $3\ \Omega$ to reference.

Solution

Nodal analysis gives $V_L = 7.00$ V and $V_R = 5.00$ V. A reading from R to L is $V_{RL} = V_R - V_L = -2.00$ V. The negative sign means L is 2.00 V above R. The resistor current is $(V_L - V_R)/4 = 0.500$ A from L to R, so it absorbs $P = (0.500)^2(4) = 1.00$ W.